Solution Requirements

Date	29 June 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/ Low)
1.	Authentication	No login/authentication required, single-user local demo application, session-based only	Low
2.	Authorization Levels	Single user role only. No differentiated access levels required for this capstone scope	Low
3.	External Interfaces	Integrates with the Wikipedia Search & Lookup API (GET requests for fact verification) and Hugging Face Transformers (local inference for DistilBERT zero-shot classification and GPT-2 Small text generation)	High
4.	Transactions Processing	No financial transactions. Each session (bio + event description + generated starters) is logged as a record written to history on generation	Medium
5.	Reporting	History view in Streamlit displays past sessions and associated feedback. No exportable reports required for this scope	Medium
6.	Business rules, etc.	A starter can only be generated once both bio and event description fields are non-empty. Fact verification runs only on themes already extracted from a generated session	High
7.	Compliance to laws or regulations	No PII beyond self-entered bio text. Data is stored locally and not transmitted to third parties beyond Wikipedia's public API, so DPDP/GDPR/HIPAA-scale controls don't apply at this scope	Low

8.	Other	System must handle a Wikipedia API miss (topic not found) gracefully, without crashing the generation flow	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric/ Acceptance Criteria
1.	Performance & Speed	Theme extraction, generation, and fact-check should each complete within a few seconds on standard hardware	Full round trip completes in under 8 seconds on local CPU inference
2.	Scalability	Designed for single local user/demo use, not concurrent multi-user load	Supports 1 concurrent local session (multi-user scaling out of scope)
3.	Security & Data Privacy	Data stored locally in JSON files only; no data transmitted except outbound GET requests to Wikipedia's public API	No external data transmission beyond Wikipedia API calls
4.	Reliability & Availability	App should not crash on empty input, network failure to Wikipedia, or a missing model file	Handles at least 3 identified failure cases without crashing
5.	Usability & Accessibility	Simple, guided Streamlit form with clear labels and placeholder text	A first-time user can generate a starter within 2 minutes with no instructions
6.	Other	Modular codebase split into event_analyzer.py, topic_generator.py, fact_checker.py for independent testing	Each module is independently unit-testable via pytest